

MUNI
FI

PB177 Cyber Attacks

Seminar 1 – Introduction to Podman, reconnaissance

Autumn 2026

Learning objectives

At the end of this lab session, you will be able to:

- set up a **virtual network environment** (sandbox) at a computer in a lab (nymfe),
- access the **sandbox** and its services via **CLI** and **GUI** from your host machine,
- use **CyberBrain**, our in-house AI assistant,
- use **nmap**, **dirb**, **generative AI**, and **web browser** for **reconnaissance** tasks.

Seminar agenda

1. Introduction to seminars
2. Containers, Podman, sandboxes
3. *“I Spy with My Network Eye”* – network enumeration using nmap
4. *“Unlisted but Not Unreachable”* – discovering directories using dirb
5. *“Did anyone teach you to phish?”* – using generative AI to our advantage
6. *“Hack It Till You Make It”* – hands-on exercises on your machines

What are these seminars for?

- **Practical, hands-on exercises** based on lecture topics
- **Introduction to common security tools** and their **basic usage**
- **Theoretical foundations**: When to use each tool, how it works, and why it matters
- **Real-world scenarios** to apply your skills **in practice**

Seminar format

- At the beginning of each seminar, there will be a short **KvIS quiz** to test your knowledge from the lectures (and earn bonus points).
- Next, you'll receive a **brief presentation** introducing the tools you'll be working with and how they function.
- For the rest of the seminar, you'll **practice** using these tools on your own machine.

Seminar organization

- **Attendance is not mandatory, but highly recommended.**
- **Completion of each lab is worth 2 points (2% of the total).**
- Bonus points towards the exam awarded from KvIS.
- No midterms.
- If you need any help, reach out to your seminar tutor, a search engine, AI or consult manual pages of tools used.
- Feedback to content is appreciated at the end of seminar.

First KvIS

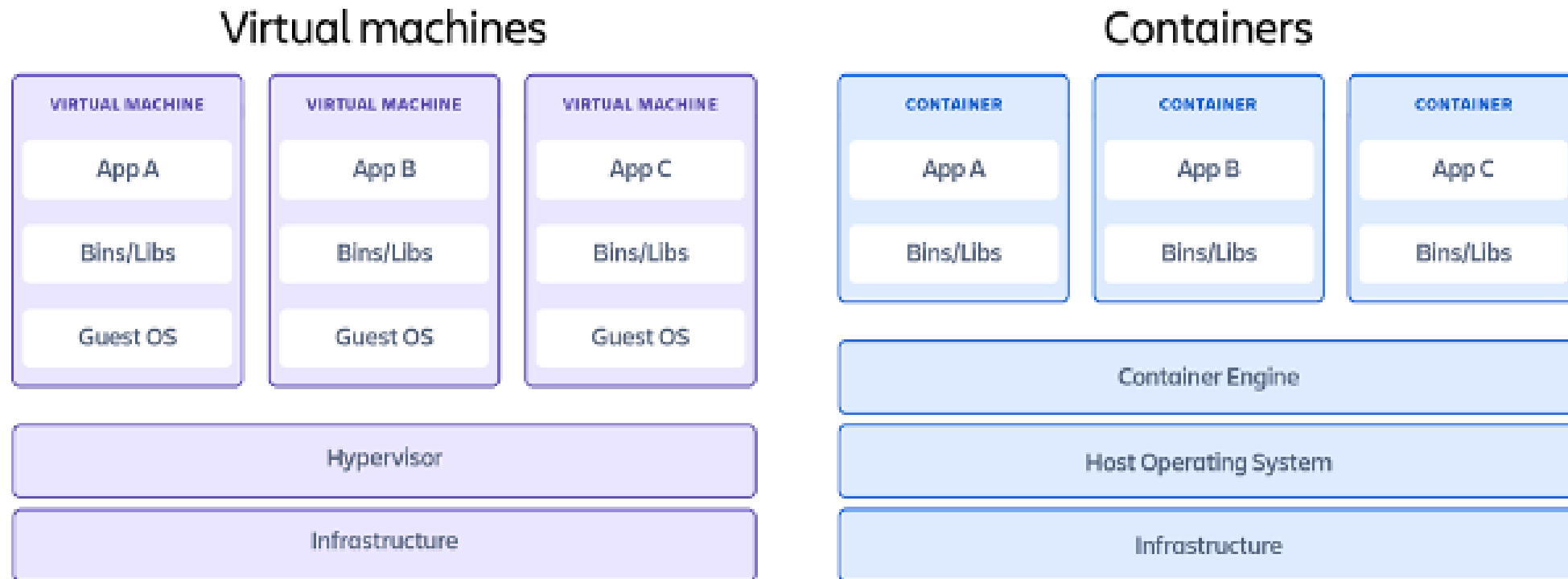
- Log into IS MU. If you are not authenticated, you cannot get bonus points.
- Go to KvIS in IS.
- Once most students are ready, the KvIS will be started.
- Nine top-performing students in all KvISes will get up to 3 bonus point for improving the final grade (see Week 1 lecture slides).
- After each KvIS, check that IS stored the points you earned.

There were a few issues last semester.

Seminar agenda

1. Introduction to seminars
- 2. Containers, Podman, sandboxes**
3. *“I Spy with My Network Eye”* – network enumeration using nmap
4. *“Unlisted but Not Unreachable”* – discovering directories using dirb
5. *“Did anyone teach you to phish?”* – using generative AI to our advantage
6. *“Hack It Till You Make It”* – hands-on exercises on your machines

Sandbox building blocks – containers



Sandbox building blocks – containers

- Containers – lightweight software packages containing all the dependencies required to execute the **contained** software applications
- Virtual machines (VMs) – heavy software packages providing complete emulation of low-level hardware devices like CPU, disk and networking devices
- In these lab sessions, we will use **containers** orchestrated by **Podman**

Podman

- Short for **pod manager**
- Open-source tool for managing containers and images
- Lightweight, rootless, daemonless
- Our **main tool** for seminars
- On your own machines, you can
use docker instead
 - Just replace every “podman” for “docker”
 - Recommended for Windows OS



Podman – your command toolkit

- **List** all running containers: `podman ps`
- List containers **forming an application**, or **sandbox** in our case:
 - Go to a directory with a **Compose** file
 - Run `podman-compose ps` there
- **Start** the application (sandbox): `podman-compose up -d`
 - **-d** means **detached** mode: Runs container **in the background**
- **Stop** the application (sandbox): `podman-compose down`
- **Stop** the application (sandbox) and **remove volumes** (disks):
`podman-compose down -v`

Preparing the sandbox – I

- Many steps **will repeat** for every seminar, homework, even for the exam
- It's good to remember these steps, but **not necessary**
 - Speeds up the process
- The steps will be provided in the following seminars too

Preparing the sandbox – II

1. Cloning/updating the Git repository with configuration files for Podman.
2. Getting access to Podman images at the faculty Gitlab.
3. Downloading the images. This will take a few minutes.
4. Logging into the INJECT Exercise platform.
5. Downloading sandbox configuration file for a particular lab.
6. Starting the sandbox and checking all containers are up and running.
7. Connecting to a container in sandbox using CLI or GUI.

Logging of shell commands

- Sandbox containers log commands entered in a shell.
- We can see your BASH and Metasploit console histories.
- We can leverage this data to help you with lab tasks as well as check the solutions to your homework.

Seminar agenda

1. Introduction to seminars
2. Containers, Podman, sandboxes
3. ***“I Spy with My Network Eye”*** – network enumeration using nmap
4. *“Unlisted but Not Unreachable”* – discovering directories using dirb
5. *“Did anyone teach you to phish?”* – using generative AI to our advantage
6. *“Hack It Till You Make It”* – hands-on exercises on your machines

Revision of Computer Networks (PB156)

- The **Application Layer** is the topmost layer in the **OSI** and **TCP/IP** models
- It **provides network services directly to users or applications** (e.g., web browsers, email clients)
- Applications use **well-known, predefined ports to communicate**, unless configured otherwise.
- The ports help **identify which application** (service) should handle incoming traffic
- Common protocols and ports:
 - HTTP/HTTPS (web) – port 80/443
 - SSH/Telnet (remote access) – ports 22/23
 - SMTP/IMAP (email) – ports 25/143
 - DNS (domain name resolution) – port 53

Network mapper – nmap

Keywords:

- port scanning
- recon
- attack vector
- exposure

– What is it?

- A powerful open-source tool for **network discovery** and **security auditing**

– What can it do?

- **Scan hosts** and **services** on a **network**
- Detect **open ports**, **running services**, and **OS versions**
- Map network topologies

– Why do we use it?

- To perform **reconnaissance** in ethical hacking
- To understand what a target system **exposes** to the network
- To identify potential **attack vectors**



Side note: cheat sheet

- Tools using command line interface (CLI), such as nmap, provide many features, so they require specifying many options.
- We crafted a [cheat sheet](#) with all tools used in the entire course.
- You may also use it during homeworks and exam!
- Make sure you **understand** the content – it's not a substitute for knowledge!

Disclaimer – Use nmap responsibly

- Nmap is a powerful tool often used for security testing — but **scanning networks without permission can have serious consequences.**
- Even a simple scan can trigger **intrusion detection systems (IDS)** or firewalls.
- University networks are **monitored** — unauthorized scanning may result in:
 - Warnings from network administrators at FI
 - An email or investigation from CSIRT-MU (Computer Security Incident Response Team)
 - Disciplinary action depending on severity

Disclaimer – Use nmap responsibly

Security incident report

Greetings,

the security team CSIRT-MU has detected involvement of the IP address [REDACTED] into the following incident:

Incident type: Anomalous Telnet traffic

Time of detection: 2024-02-16 11:31:44 +0100

IP address: [REDACTED]

Domain name: student-152.eduroam.muni.cz

Details of the incident

Detailed information about this incident can be found on our [security portal \[3\]](#).

Description of the incident

Computer with the aforementioned IP address was excessively communicating using the Telnet protocol. This protocol is rarely used nowadays. This activity most often means that the computer is infected by a virus or other malicious code, or runs some misconfigured service.

Incident solution

We strongly recommend to check the computer with up-to-date antivirus software and eventually check the configuration of network services.

[How to communicate with the CSIRT-MU team? \[4\]](#)

Please review the source of the incident and fix the issue. Inform us about the result within 5 business days.

Best regards,

CSIRT-MU

The security team of Masaryk University

<https://csirt.muni.cz>

Seminar agenda

1. Introduction to seminars
2. Containers, Podman, sandboxes
3. *“I Spy with My Network Eye”* – network enumeration using nmap
4. ***“Unlisted but Not Unreachable”*** – discovering directories using dirb
5. *“Did anyone teach you to phish?”* – using generative AI to our advantage
6. *“Hack It Till You Make It”* – hands-on exercises on your machines

Web Content Scanner – dirb

Keywords:

- subdirectories
- brute force
- wordlist

– What is it?

- A command-line tool that **brute-forces web URLs** using **wordlists**

– What can it do?

- Finds **hidden directories and files** (e.g., /admin, /backup.zip) not linked on websites

– Why do we use it?

- To **uncover potential entry points** for further testing and **detect misconfigurations** in web apps
- Always ensure you have **proper authorization** before scanning to avoid **legal issues** or triggering **security alerts**



Other similar tools

- enum4linux – SMB and Windows info enumeration
- wpscan – WordPress vulnerability scanner
- rustscan – Fast port scanner, built in Rust
- gobuster – Fast brute-forcer for web dirs, DNS, and virtual hosts

Seminar agenda

1. Introduction to seminars
2. Containers, Podman, sandboxes
3. *“I Spy with My Network Eye”* – network enumeration using nmap
4. *“Unlisted but Not Unreachable”* – discovering directories using dirb
5. ***“Did anyone teach you to phish?”*** – using generative AI to our advantage
6. *“Hack It Till You Make It”* – hands-on exercises on your machines

AI-Powered Phishing

- Generative AI (like ChatGPT) can create **natural, convincing, and error-free** phishing messages.
- How it is used:
 - Mimicking **corporate language** or brands
 - Generating **personalized messages** based on public data (e.g., LinkedIn)
 - **Localizing language** for different regions
 - Creating **fake login pages** or SMS content
- Why it is a threat:
 - Scales easily – thousands of variations in **seconds**
 - Harder to spot – **no spelling errors**, fluent wording
 - Bypasses filters – randomization and subtlety **reduce detection**

AI-Generated Email Example

Subject: 🚨 Urgent: Account Access Suspended

Dear [Your Name],

We detected suspicious activity on your university account and have temporarily suspended access to prevent unauthorized use.

To restore access, please verify your identity by logging in at the secure portal below:

👉 [Restore Access Now](#)

If you do not act within 24 hours, your account may be permanently locked.

Thank you for helping us keep your information safe.

— IT Support Team

CyberBrain

- An **in-house AI assistant with open-weight LLMs and knowledge base of course materials.**
- Feel free to use it when solving labs and homeworks.
- However, **do not delegate the work to assistants**, use them to **support your learning.**
- You will get a limited access to it at the exam.

Cybersecurity educational research

- Cybersecurity lab run research on how students use platforms to learn cybersecurity and on using AI assistants when solving cybersecurity tasks.
- We ask you for permission for analyzing data about your interactions with the INJECT platform, sandbox shell commands, and CyberBrain prompts.
- **If you would like to help us and advance how cybersecurity is taught,** please provide us your consent in the dialog in the INJECT platform.
- The data collected data will be **anonymized**.
- Participation in the research is **voluntary**.

Seminar agenda

1. Introduction to seminars
2. Containers, Podman, sandboxes
3. *“I Spy with My Network Eye”* – network enumeration using nmap
4. *“Unlisted but Not Unreachable”* – discovering directories using dirb
5. *“Did anyone teach you to phish?”* – using generative AI to our advantage
6. ***“Hack It Till You Make It”*** – hands-on exercises on your machines

Now that you know your toolkit...

- It's time to put **theory** into **practice**!
- In the following part of the seminar, you'll **try out the tools** you've just learned about.
- You'll explore **basic usage**, **test real-world scenarios**, and **get a feel** for how these tools **behave in action**
- Don't worry if you make mistakes, this is where learning really begins.

Lab in the INJECT Platform

- Log into the INJECT Exercise platform at <https://inject3.fi.muni.cz/> using the credentials sent to your university email UČO@mail.muni.cz.
- Enter the lab exercise in the platform.
- The platform will gradually display tasks that you will complete in your sandbox.
- Work at your own pace.
- Feel free to show the solution on the platform, use “Get help” for each task if needed, ask CyberBrain or the tutor.
- If you do not complete the lab in class, complete it at home in 2 weeks.
- After that, we will check whether you finished and award points for completion.

Adjust INJECT to fit your screen

- Light and dark modes available  
- Change container width to fully utilize your big screen in Settings.

Go to Settings of your account (top right)

Container width

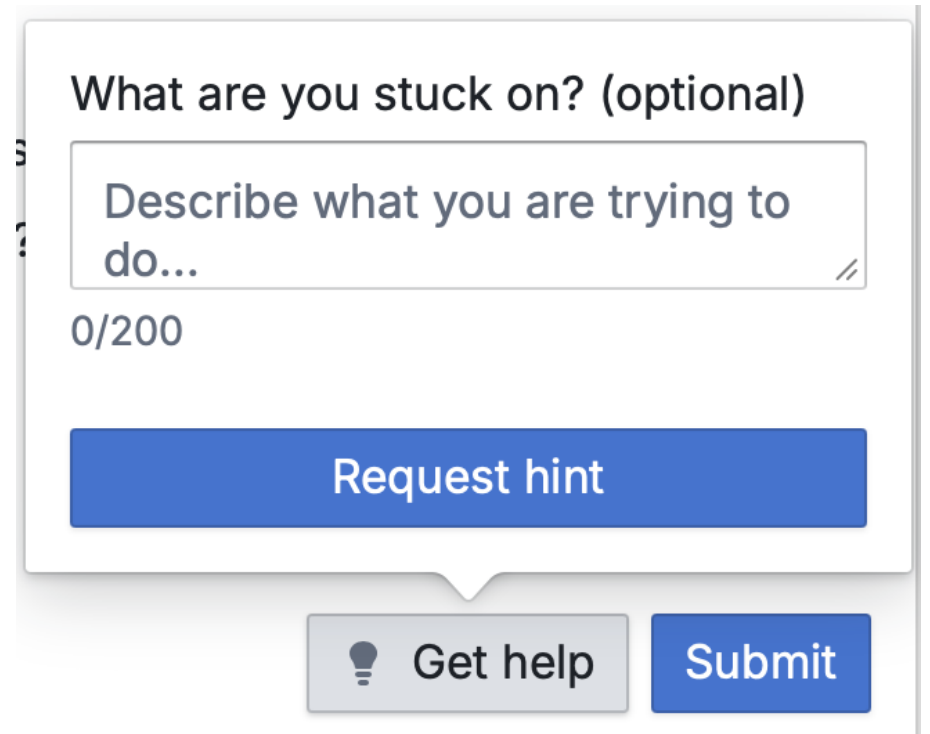
Set the maximum width of the main container in rem (multiples of font size). This setting can help you optimize the layout for your screen size.



Get help feature

- Each question in INJECT contains **Get help** button.
- This is a brand-new feature.
- Help us with its testing and if you use it, please rate the answer.

Was this hint helpful?  



What are you stuck on? (optional)

Describe what you are trying to do...

0/200

Request hint

 Get help Submit

Note where you are seating

- Please use the same nymfe next time to save disk space
- Make a note or take a photo of the nymfe number, such as
nymfe89

MUNI
FI